

Software Requirements Specification (SRS)

WordPress-Style Drag & Drop Website Builder

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Version	Date	Author	Description
1.0	Dec 6, 2024	Development Team	Initial SRS Document

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a complete description of all functions and specifications of the Visual Website Builder application. This document is intended for:

- Stakeholders:** To understand the system capabilities and limitations
- Developers:** To implement the system according to specifications

- **Testers:** To develop test cases and validation criteria
- **Project Managers:** To track project progress and requirements coverage
- **End Users:** To understand system capabilities and features

1.2 Scope

Product Name: Visual Website Builder

Product Description:

The Visual Website Builder is a web-based application that enables users to create professional static websites through an intuitive drag-and-drop interface without requiring programming knowledge. The system allows users to visually compose web pages using pre-built components, edit content inline, preview designs, and export fully functional HTML files.

Goals:

- Democratize web development for non-technical users
- Reduce website creation time from days to minutes
- Generate clean, standards-compliant HTML/CSS code
- Provide WordPress-like editing experience in the browser
- Enable rapid prototyping and landing page creation

Benefits:

- Zero coding knowledge required
- No server-side dependencies
- Instant preview and export
- Professional-looking results
- Cost-effective website creation

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
UI	User Interface
UX	User Experience
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
DOM	Document Object Model
Component	Reusable UI building block (header, text, image, etc.)
Canvas	Main editing area where components are placed

Artifact	Final exported HTML file
Drag & Drop	Interaction method for moving components
WYSIWYG	What You See Is What You Get
Static Site	Website with fixed content (no server-side processing)
SPA	Single Page Application
API	Application Programming Interface
Responsive Design	Design that adapts to different screen sizes

1.4 References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
- W3C HTML5 Specification
- W3C CSS3 Specification
- React Documentation (v18.x)
- WCAG 2.1 (Web Content Accessibility Guidelines)
- Material Design Guidelines

1.5 Overview

This document is organized into seven main sections:

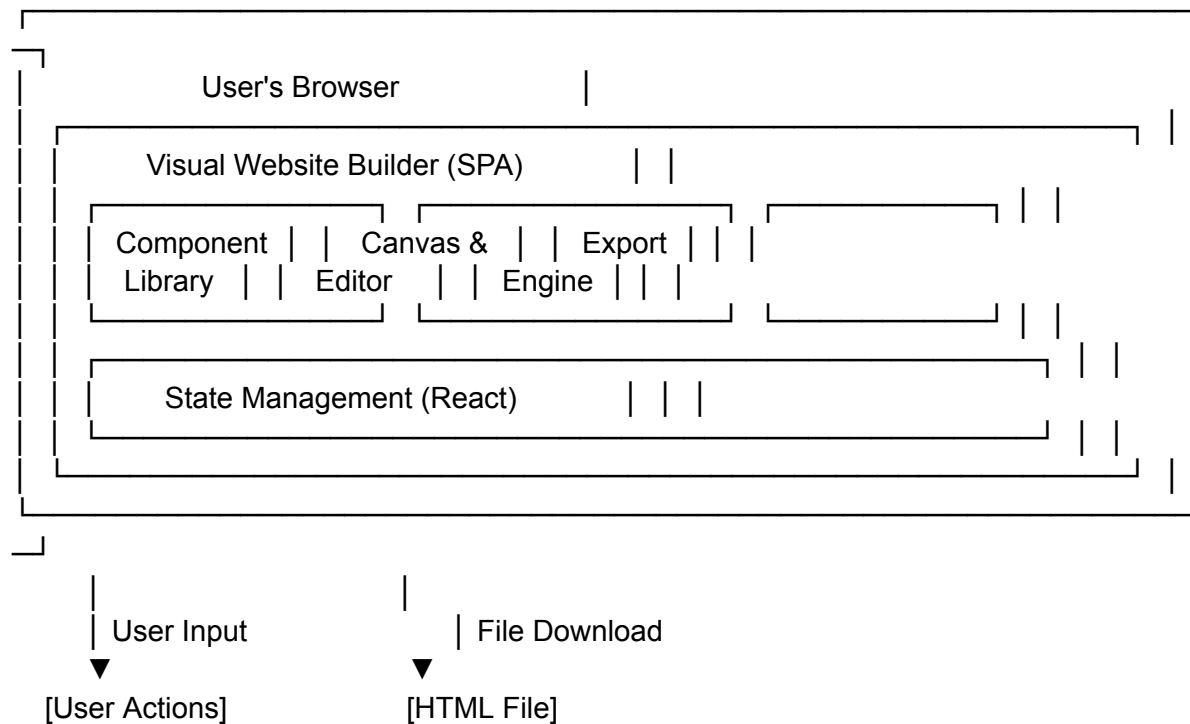
- **Section 1:** Introduction and document overview
- **Section 2:** General description of the system and context
- **Section 3:** Detailed functional and system requirements
- **Section 4:** External interface specifications
- **Section 5:** Non-functional requirements (performance, security, etc.)
- **Section 6:** Additional requirements and constraints
- **Section 7:** Appendices and supporting information

2. Overall Description

2.1 Product Perspective

The Visual Website Builder is a standalone web application that operates entirely within the user's browser. It does not require backend services, databases, or user authentication for basic functionality.

2.1.1 System Context



2.1.2 System Interfaces

- **Browser APIs:** DOM manipulation, File Download API, Drag & Drop API
- **Storage:** Browser LocalStorage (optional for save/load)
- **No external dependencies:** Completely client-side application

2.1.3 Product Functions Overview

1. Component Management

- Add pre-built components to canvas
- Delete components
- Reorder components via drag & drop
- Duplicate components

2. Content Editing

- Inline editing of text content
- Image URL configuration
- Button link configuration
- Multi-column layout editing

3. Preview System

- Real-time preview mode
- WYSIWYG editing experience
- View final output before export

4. Export System

- Generate standalone HTML file
- Embedded CSS styling
- View generated source code
- Download functionality

2.2 Product Functions

2.2.1 Component Library

- **FR-CL-01:** System shall provide 5+ pre-built component types
- **FR-CL-02:** Each component shall have default content and styling
- **FR-CL-03:** Components shall be accessible via left sidebar panel
- **FR-CL-04:** Components shall be added via single click

2.2.2 Canvas Editor

- **FR-CE-01:** System shall provide main editing canvas for component placement
- **FR-CE-02:** Canvas shall support unlimited components
- **FR-CE-03:** Components shall be visually distinct and separated
- **FR-CE-04:** Empty canvas shall display helpful onboarding message

2.2.3 Drag & Drop System

- **FR-DD-01:** Users shall reorder components via drag & drop
- **FR-DD-02:** System shall provide visual feedback during drag operation
- **FR-DD-03:** Components shall snap into new positions on drop
- **FR-DD-04:** Drag handle shall be clearly visible on each component

2.2.4 Editing System

- **FR-ED-01:** Users shall edit component content inline
- **FR-ED-02:** Edit mode shall be activated via edit button
- **FR-ED-03:** Changes shall be saved automatically
- **FR-ED-04:** Edit mode shall provide appropriate input fields per component type

2.2.5 Preview System

- **FR-PR-01:** System shall provide full-screen preview mode
- **FR-PR-02:** Preview shall render exact final output
- **FR-PR-03:** Users shall exit preview to return to editing
- **FR-PR-04:** Preview shall hide editing controls

2.2.6 Export System

- **FR-EX-01:** System shall generate valid HTML5 documents
- **FR-EX-02:** Generated HTML shall include embedded CSS
- **FR-EX-03:** Users shall download HTML via browser download
- **FR-EX-04:** System shall provide code view of generated HTML

2.3 User Characteristics

2.3.1 Primary User Personas

Persona 1: Small Business Owner

- Age: 30-50 years
- Technical Skill: Low to Medium
- Goal: Create simple business website quickly
- Pain Points: Limited budget, no coding skills
- Frequency of Use: Occasional (during business setup or updates)

Persona 2: Marketing Professional

- Age: 25-40 years
- Technical Skill: Medium
- Goal: Create landing pages for campaigns rapidly
- Pain Points: Dependency on developers, slow turnaround
- Frequency of Use: Frequent (weekly campaign launches)

Persona 3: Freelance Designer

- Age: 22-35 years
- Technical Skill: Medium to High
- Goal: Prototype client websites quickly
- Pain Points: Time-consuming development, client iterations
- Frequency of Use: Very Frequent (daily prototyping)

Persona 4: Student/Educator

- Age: 18-60 years
- Technical Skill: Low to Medium
- Goal: Create portfolio or educational content
- Pain Points: Learning curve of traditional tools
- Frequency of Use: Occasional (project-based)

2.3.2 User Skill Requirements

- **Minimum:** Basic computer literacy, web browsing skills
- **Recommended:** Understanding of basic web concepts (links, images)
- **Not Required:** Programming knowledge, HTML/CSS expertise

2.4 Constraints

2.4.1 Technical Constraints

- **TC-01:** Application must run entirely in browser (no server required)
- **TC-02:** Must support modern browsers (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+)
- **TC-03:** Must use React framework for UI development

- **TC-04:** Must not require user authentication for basic features
- **TC-05:** File size of exported HTML must be under 5MB for practical use

2.4.2 Business Constraints

- **BC-01:** Development must be completed within 12 weeks
- **BC-02:** Must be deployable on static hosting platforms
- **BC-03:** Must be free to use (no licensing costs for end users)
- **BC-04:** Must work without internet connection after initial load

2.4.3 Regulatory Constraints

- **RC-01:** Must comply with WCAG 2.1 Level AA accessibility standards
- **RC-02:** Generated HTML must comply with W3C HTML5 standards
- **RC-03:** Must respect user privacy (no data collection without consent)
- **RC-04:** Must include proper copyright and licensing information

2.5 Assumptions and Dependencies

2.5.1 Assumptions

- **AS-01:** Users have access to modern web browsers
- **AS-02:** Users have stable internet connection for initial load
- **AS-03:** Users understand basic web concepts (URLs, images)
- **AS-04:** Users can download files from browser
- **AS-05:** Users have basic mouse/touch interaction capabilities

2.5.2 Dependencies

- **DP-01:** React library (v18.x or higher)
- **DP-02:** Lucide React icon library
- **DP-03:** Modern browser with ES6+ support
- **DP-04:** Browser Drag & Drop API support
- **DP-05:** Browser File Download API support

3. System Features and Requirements

3.1 Component Management System

Priority: High

Description: Core system for managing page components

3.1.1 Functional Requirements

FR-3.1.1: Component Addition

- **ID:** FR-CMS-001

- **Description:** User shall be able to add components from component library
- **Input:** Click on component template in sidebar
- **Process:** Create new component instance with default content
- **Output:** Component appears at bottom of canvas
- **Priority:** Critical

FR-3.1.2: Component Deletion

- **ID:** FR-CMS-002
- **Description:** User shall be able to delete any component
- **Input:** Click delete button on component
- **Process:** Remove component from canvas
- **Output:** Component removed, canvas updates
- **Priority:** Critical

FR-3.1.3: Component Reordering

- **ID:** FR-CMS-003
- **Description:** User shall reorder components via drag & drop
- **Input:** Drag component to new position
- **Process:** Update component order in state
- **Output:** Component moved to new position
- **Priority:** High

FR-3.1.4: Component Identification

- **ID:** FR-CMS-004
- **Description:** Each component shall have unique identifier
- **Input:** Component creation
- **Process:** Generate timestamp-based unique ID
- **Output:** Component with unique ID
- **Priority:** Critical

3.1.2 Component Types

FR-3.1.5: Header Component

- **ID:** FR-CT-001
- **Description:** Hero section with title and subtitle
- **Fields:** title (string), subtitle (string)
- **Default:** "Welcome to My Website", "Your tagline here"
- **Styling:** Gradient background, centered text, large typography

FR-3.1.6: Text Block Component

- **ID:** FR-CT-002
- **Description:** Paragraph text content block
- **Fields:** text (string, multiline)
- **Default:** "Add your content here. Click edit to customize."
- **Styling:** Standard paragraph formatting, readable line height

FR-3.1.7: Image Component

- **ID:** FR-CT-003
- **Description:** Image display with optional caption
- **Fields:** url (string), alt (string), caption (string, optional)
- **Default:** Placeholder image URL
- **Styling:** Centered, responsive, with shadow

FR-3.1.8: Button Component

- **ID:** FR-CT-004
- **Description:** Call-to-action button with link
- **Fields:** text (string), link (string/URL)
- **Default:** "Click Me", "#"
- **Styling:** Prominent button, hover effects

FR-3.1.9: Two-Column Section Component

- **ID:** FR-CT-005
- **Description:** Side-by-side content layout
- **Fields:** col1 (string, multiline), col2 (string, multiline)
- **Default:** "Column 1 content", "Column 2 content"
- **Styling:** Grid layout, equal width columns, responsive

3.2 Content Editing System

Priority: High

Description: System for inline content editing

3.2.1 Functional Requirements

FR-3.2.1: Edit Mode Activation

- **ID:** FR-EDIT-001
- **Description:** User shall activate edit mode for any component
- **Input:** Click edit button on component
- **Process:** Toggle edit state for component
- **Output:** Edit fields displayed, content editable
- **Priority:** Critical

FR-3.2.2: Real-time Content Update

- **ID:** FR-EDIT-002
- **Description:** Content changes shall update immediately
- **Input:** User types in edit field
- **Process:** Update component state on each change
- **Output:** Content updated in real-time
- **Priority:** High

FR-3.2.3: Edit Field Validation

- **ID:** FR-EDIT-003
- **Description:** System shall validate user input
- **Input:** User enters content
- **Process:** Validate against field requirements
- **Output:** Accepted or error message
- **Priority:** Medium

FR-3.2.4: Edit Mode Exit

- **ID:** FR-EDIT-004
- **Description:** User shall exit edit mode
- **Input:** Click edit button again or click outside
- **Process:** Exit edit mode, save changes
- **Output:** Component displays updated content
- **Priority:** High

3.2.2 Input Field Specifications

FR-3.2.5: Text Input Fields

- **Type:** Single-line text input
- **Max Length:** 500 characters
- **Validation:** No HTML tags allowed
- **Placeholder:** Context-appropriate placeholder text

FR-3.2.6: Textarea Fields

- **Type:** Multi-line text input
- **Max Length:** 5000 characters
- **Min Height:** 128px
- **Auto-resize:** Optional

FR-3.2.7: URL Input Fields

- **Type:** Single-line text input
- **Validation:** Valid URL format (optional validation)
- **Placeholder:** "https://example.com"

3.3 Drag & Drop System

Priority: Medium

Description: Visual reordering of components

3.3.1 Functional Requirements

FR-3.3.1: Drag Initiation

- **ID:** FR-DND-001
- **Description:** User shall initiate drag on component
- **Input:** Mouse down on drag handle

- **Process:** Set draggable state
- **Output:** Component becomes draggable
- **Priority:** High

FR-3.3.2: Drag Visual Feedback

- **ID:** FR-DND-002
- **Description:** System shall provide visual feedback during drag
- **Input:** Drag in progress
- **Process:** Apply visual styling to dragged component
- **Output:** User sees component being dragged
- **Priority:** Medium

FR-3.3.3: Drop Target Indication

- **ID:** FR-DND-003
- **Description:** System shall indicate valid drop locations
- **Input:** Drag over potential drop location
- **Process:** Highlight drop target
- **Output:** Visual indicator of drop location
- **Priority:** Low

FR-3.3.4: Component Reordering

- **ID:** FR-DND-004
- **Description:** Components shall reorder on drop
- **Input:** Drop component at new position
- **Process:** Update component order in state
- **Output:** Components reordered on canvas
- **Priority:** High

3.4 Preview System

Priority: High

Description: Full-screen preview of final output

3.4.1 Functional Requirements

FR-3.4.1: Preview Mode Activation

- **ID:** FR-PREV-001
- **Description:** User shall activate preview mode
- **Input:** Click "Preview" button
- **Process:** Switch to preview mode
- **Output:** Full-screen preview displayed
- **Priority:** High

FR-3.4.2: Preview Rendering

- **ID:** FR-PREV-002

- **Description:** Preview shall render exact final output
- **Input:** Current component state
- **Process:** Render components without editing controls
- **Output:** WYSIWYG preview
- **Priority:** Critical

FR-3.4.3: Preview Exit

- **ID:** FR-PREV-003
- **Description:** User shall exit preview mode
- **Input:** Click "Exit Preview" button
- **Process:** Return to editing mode
- **Output:** Editor displayed with all controls
- **Priority:** High

FR-3.4.4: Preview Toolbar

- **ID:** FR-PREV-004
- **Description:** Preview shall display minimal toolbar
- **Input:** Preview mode active
- **Process:** Show toolbar with exit button
- **Output:** Fixed toolbar at top of screen
- **Priority:** Medium

3.5 Export System

Priority: Critical

Description: Generate and download HTML files

3.5.1 Functional Requirements

FR-3.5.1: HTML Generation

- **ID:** FR-EXP-001
- **Description:** System shall generate valid HTML5 document
- **Input:** Current component state
- **Process:** Convert components to HTML markup
- **Output:** Complete HTML document as string
- **Priority:** Critical

FR-3.5.2: CSS Embedding

- **ID:** FR-EXP-002
- **Description:** Generated HTML shall include embedded CSS
- **Input:** Component styles
- **Process:** Generate CSS rules, embed in `<style>` tag
- **Output:** HTML with embedded styles
- **Priority:** Critical

FR-3.5.3: File Download

- **ID:** FR-EXP-003
- **Description:** User shall download generated HTML
- **Input:** Click "Download" button
- **Process:** Create blob, trigger browser download
- **Output:** HTML file downloaded to user's device
- **Priority:** Critical

FR-3.5.4: Code View

- **ID:** FR-EXP-004
- **Description:** User shall view generated source code
- **Input:** Click "View Code" button
- **Process:** Display HTML in code viewer
- **Output:** Formatted HTML displayed with syntax highlighting
- **Priority:** Medium

FR-3.5.5: File Naming

- **ID:** FR-EXP-005
- **Description:** Downloaded file shall have descriptive name
- **Input:** Download action
- **Process:** Generate filename
- **Output:** File named "website.html" (or custom name)
- **Priority:** Low

3.5.2 HTML Generation Specifications

FR-3.5.6: HTML Structure

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Website</title>
  <style>/* CSS here */</style>
</head>
<body>
  <!-- Components here -->
</body>
</html>
```

FR-3.5.7: CSS Requirements

- Reset/normalize styles included
- Responsive breakpoints defined

- Component-specific classes
- Hover/focus states
- Mobile-first approach

FR-3.5.8: Validation

- Generated HTML must pass W3C validator
- No broken tags or unclosed elements
- Proper nesting and semantics
- Valid CSS syntax

3.6 User Interface System

Priority: High

Description: Overall UI layout and navigation

3.6.1 Functional Requirements

FR-3.6.1: Top Toolbar

- **ID:** FR-UI-001
- **Description:** Fixed toolbar with primary actions
- **Location:** Top of screen, sticky positioned
- **Contents:** App title, Preview, View Code, Download buttons
- **Priority:** High

FR-3.6.2: Component Sidebar

- **ID:** FR-UI-002
- **Description:** Left sidebar with component library
- **Location:** Left side, sticky positioned
- **Contents:** Component templates with add buttons
- **Width:** 25% of viewport (3 columns in 12-column grid)
- **Priority:** High

FR-3.6.3: Main Canvas

- **ID:** FR-UI-003
- **Description:** Primary editing area for components
- **Location:** Right side of sidebar
- **Contents:** Added components with controls
- **Width:** 75% of viewport (9 columns in 12-column grid)
- **Priority:** Critical

FR-3.6.4: Component Controls

- **ID:** FR-UI-004
- **Description:** Control bar on each component
- **Location:** Top of component
- **Contents:** Drag handle, type label, edit button, delete button

- **Priority:** High

FR-3.6.5: Empty State

- **ID:** FR-UI-005
 - **Description:** Helpful message when canvas is empty
 - **Contents:** Icon, title, description
 - **Action:** Encourages user to add first component
 - **Priority:** Low
-

4. External Interface Requirements

4.1 User Interface Requirements

4.1.1 General UI Requirements

UIR-01: Responsive Design

- Interface shall adapt to screen sizes 1024px and above
- Tablet mode (768px-1023px): Adjusted layout
- Mobile warning for screens below 768px
- Touch-friendly controls (minimum 44x44px tap targets)

UIR-02: Visual Design

- Modern, clean, minimal aesthetic
- Consistent color scheme (primary: purple, secondary: gray)
- Adequate whitespace and padding
- Clear visual hierarchy
- Accessible color contrast ratios (WCAG AA minimum)

UIR-03: Typography

- System font stack for performance
- Readable font sizes (minimum 16px for body text)
- Appropriate line heights (1.5-1.8 for body text)
- Clear heading hierarchy

UIR-04: Interactive Elements

- Buttons shall have hover and active states
- Focus states for keyboard navigation
- Loading indicators for async operations
- Smooth transitions (200-300ms)

4.1.2 Specific UI Components

Component Library Panel

- Clear component icons (20px size)
- Component labels
- Add button (+) on each template
- Hover effects on templates
- Scrollable if content overflows

Canvas Area

- Clear component boundaries
- Visual separation between components
- Component control bar always visible
- Drag handles clearly indicated
- Edit/delete buttons with appropriate icons

Preview Mode

- Full-screen display
- Minimal toolbar (exit button only)
- Exact replication of final output
- Dark toolbar to distinguish from content

Code View

- Monospace font for code display
- Syntax highlighting (optional)
- Scrollable code block
- Copy button for code (optional)
- Back button to return to editor

4.2 Hardware Interfaces

HIR-01: No specialized hardware required beyond standard computer input devices

HIR-02: Mouse or touchpad for drag & drop operations (recommended)

HIR-03: Keyboard for text input

HIR-04: Touch screen support (optional, for mobile/tablet devices)

4.3 Software Interfaces

4.3.1 Browser APIs

SWI-01: DOM API

- **Purpose:** Manipulate document structure
- **Version:** DOM Level 4
- **Usage:** Component rendering, event handling

SWI-02: File API

- **Purpose:** Generate and download files
- **Methods:** Blob(), URL.createObjectURL(), createElement('a')
- **Usage:** HTML file export

SWI-03: Drag and Drop API

- **Purpose:** Enable drag & drop functionality
- **Events:** dragstart, dragenter, dragover, drop, dragend
- **Usage:** Component reordering

SWI-04: Local Storage API (Optional)

- **Purpose:** Save/load projects locally
- **Methods:** localStorage.setItem(), localStorage.getItem()
- **Usage:** Project persistence

4.3.2 External Libraries

SWI-05: React (v18.x)

- **Purpose:** UI framework
- **Components:** useState, useRef, useEffect hooks
- **Usage:** Component-based architecture, state management

SWI-06: Lucide React

- **Purpose:** Icon library
- **Icons:** Plus, Trash2, Edit, Download, Eye, Code, GripVertical, Type, Image, Layout, Square
- **Usage:** UI iconography

4.3.3 No Backend Services

- System operates entirely client-side
- No API calls required for basic functionality
- No authentication or user management
- No database connections

4.4 Communication Interfaces

COM-01: No network communication required after initial page load

COM-02: Optional: Future enhancement for cloud save features

COM-03: Downloaded HTML files are standalone (no external dependencies)

5. Non-Functional Requirements

5.1 Performance Requirements

NFR-PERF-01: Load Time

- Initial application load: < 3 seconds on standard broadband
- Component library shall load instantly
- No perceived lag in UI interactions

NFR-PERF-02: Responsiveness

- UI interactions shall respond within 100ms
- Drag & drop operations shall be smooth (60fps target)
- Component additions shall be instantaneous
- Edit mode activation: < 50ms

NFR-PERF-03: Component Limits

- System shall support minimum 100 components on canvas
- No performance degradation up to 50 components
- Graceful performance reduction for 50-100 components
- Warning if approaching performance limits

NFR-PERF-04: Export Performance

- HTML generation: < 1 second for typical page (20 components)
- Download initiation: < 500ms
- File size: < 500KB for typical page

NFR-PERF-05: Memory Usage

- Application memory footprint: < 100MB
- No memory leaks during extended use
- Garbage collection handled appropriately

5.2 Security Requirements

NFR-SEC-01: Client-Side Security

- No user authentication required (reduces attack surface)
- No sensitive data storage
- All operations performed client-side
- No data transmission to external servers

NFR-SEC-02: Input Sanitization

- User input shall be escaped in HTML output
- Protection against XSS in generated HTML

- URL validation for link fields
- Image URL validation

NFR-SEC-03: Generated Code Security

- No inline JavaScript in generated HTML
- No external script dependencies
- Safe CSS (no JavaScript URLs)
- Content Security Policy friendly

NFR-SEC-04: Privacy

- No user tracking or analytics (unless explicitly added)
- No cookies or session storage
- No PII collection
- Local-only operation

5.3 Reliability Requirements

NFR-REL-01: Error Handling

- Graceful handling of invalid inputs
- User-friendly error messages
- No application crashes or freezes
- Automatic recovery from errors when possible

NFR-REL-02: Data Integrity

- Component state maintained accurately
- No data loss during editing operations
- Undo capability preserves state history (future enhancement)
- Export generates accurate representation

NFR-REL-03: Availability

- Application available 100% when user has browser open
- No downtime for maintenance (client-side only)
- Works offline after initial load (with service worker)

NFR-REL-04: Fault Tolerance

- System continues operating if single component fails
- Corrupted component can be deleted
- Browser refresh recovers clean state

5.4 Usability Requirements

NFR-USE-01: Learnability

- New users shall create basic page within 5 minutes

- No tutorial required for basic operations
- Intuitive interface following platform conventions
- Tooltips and helper text where appropriate

NFR-USE-02: Efficiency

- Experienced users shall create page in < 2 minutes
- Keyboard shortcuts for common actions (future)
- Minimal clicks for common operations
- Quick access to all features

NFR-USE-03: Memorability

- Users shall remember how to use after 1 week absence
- Consistent interaction patterns
- Clear labeling of all features
- Familiar metaphors (drag & drop, etc.)

NFR-USE-04: Error Prevention

- Confirmation for destructive actions (delete)
- Clear labeling of all buttons and controls
- Disabled states for unavailable actions
- Visual feedback for all interactions

NFR-USE-05: Satisfaction

- Pleasant visual design
- Smooth animations and transitions
- Immediate feedback for actions
- Sense of accomplishment on completion

5.5 Maintainability Requirements

NFR-MAINT-01: Code Quality

- Modular architecture with separation of concerns
- Component-based design
- Clear function naming and documentation
- Consistent code style

NFR-MAINT-02: Extensibility

- Easy to add new component types
- Plugin architecture for future enhancements
- Minimal coupling between modules
- Well-defined interfaces

NFR-MAINT-03: Documentation

- Inline code comments for complex logic
- API documentation for key functions
- User documentation and tutorials
- Deployment instructions

NFR-MAINT-04: Testing

- Unit tests for critical functions
- Integration tests for workflows
- Manual testing checklist
- Browser compatibility testing

5.6 Portability Requirements

NFR-PORT-01: Browser Support

- Chrome 90+ (priority: high)
- Firefox 88+ (priority: high)
- Safari 14+ (priority: medium)
- Edge 90+ (priority: medium)

NFR-PORT-02: Operating System Support

- Windows 10/11
- macOS 10.15+
- Linux (major distributions)
- iOS 14+ (limited support)
- Android 10+ (limited support)

NFR-PORT-03: Deployment

- Deployable on any static web host
- No server-side requirements
- Single page application architecture
- CDN-friendly

5.7 Accessibility Requirements

NFR-ACC-01: WCAG 2.1 Level AA Compliance

- Color contrast ratios minimum 4.5:1 for normal text
- Color contrast ratios minimum 3:1 for large text
- All functionality available via keyboard
- Focus indicators clearly visible

NFR-ACC-02: Screen Reader Support

- Semantic HTML in generated output
- ARIA labels where appropriate

- Logical tab order
- Descriptive link text

NFR-ACC-03: Visual Accommodations

- Support for browser zoom up to 200%
- No information conveyed by color alone
- Clear visual hierarchy
- Adequate spacing between interactive elements

NFR-ACC-04: Cognitive Accommodations

- Simple, clear language
- Consistent navigation
- No time limits on user actions
- Avoidance of flashing content

5.8 Scalability Requirements

NFR-SCAL-01: User Scalability

- Support unlimited concurrent users (client-side)
- No server bottlenecks
- No user data conflicts (no shared state)

NFR-SCAL-02: Data Scalability

- Support pages up to 100 components
- Efficient state management
- Lazy loading for large projects (future)

NFR-SCAL-03: Feature Scalability

- Architecture supports adding new component types
- Extensible export system
- Plugin system for future enhancements

6. Other Requirements

6.1 Legal Requirements

LR-01: Licensing

- Application code licensed under MIT License
- Third-party library licenses respected
- Generated HTML belongs to user (no license restrictions)

LR-02: Intellectual Property

- No trademarked terms in generated content
- No copyrighted content in default templates
- User responsible for content legality

LR-03: Privacy Policy

- Clear statement of no data collection
- Explanation of local-only operation
- User consent for any future analytics

LR-04: Terms of Service

- No warranty of fitness for particular purpose
- User responsibility for generated content
- Acceptable use policy

6.2 Standards Compliance

SC-01: Web Standards

- HTML5 specification compliance
- CSS3 specification compliance
- ECMAScript 2015+ (ES6+) for application code
- W3C validation for generated output

SC-02: Accessibility Standards

- WCAG 2.1 Level AA compliance
- Section 508 compliance
- ARIA 1.2 specification

SC-03: Code Quality Standards

- ESLint for JavaScript linting
- Prettier for code formatting
- Consistent naming conventions
- Code review process

6.3 Internationalization Requirements

I18N-01: Language Support

- Application interface in English (v1.0)
- Architecture supports future translations
- UTF-8 character encoding throughout
- Support for right-to-left languages (future)

I18N-02: Regional Formats

- Date/time formats (if used) follow locale
- Number formats follow locale
- Currency formats (if used) follow locale

6.4 Environmental Requirements

ENV-01: Browser Requirements

- Modern browser with JavaScript enabled
- Minimum screen resolution: 1024x768
- Recommended screen resolution: 1920x1080
- Mouse or touchpad (recommended)

ENV-02: Network Requirements

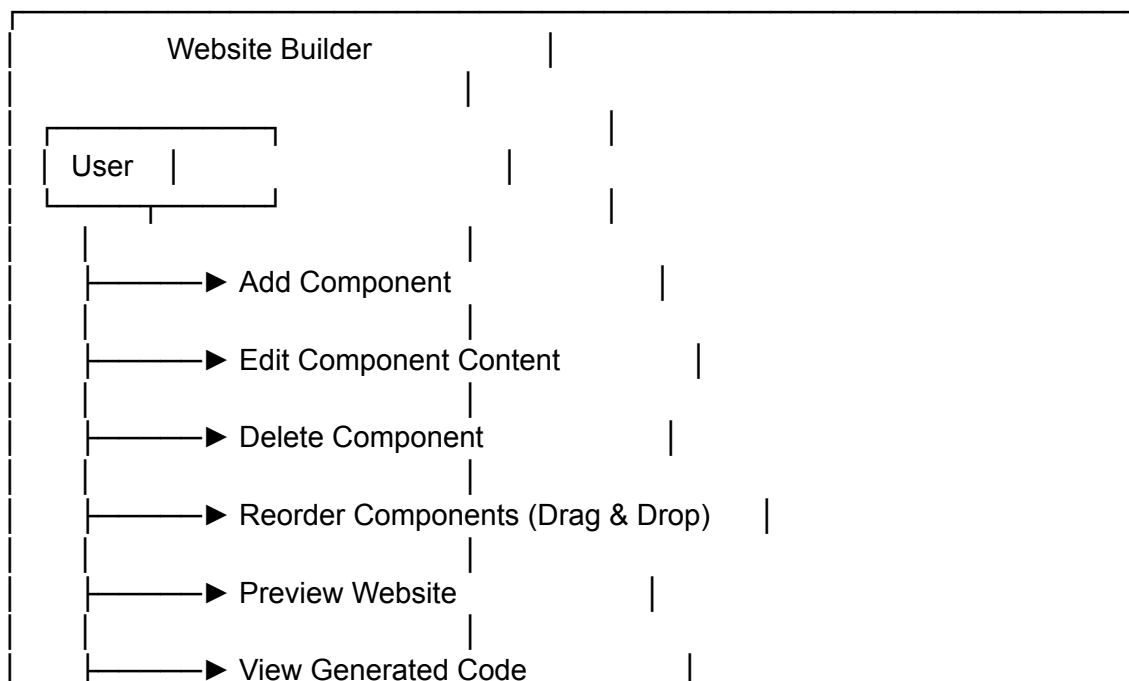
- Internet connection for initial load
- Offline functionality after load (with service worker)
- No continuous network connection required

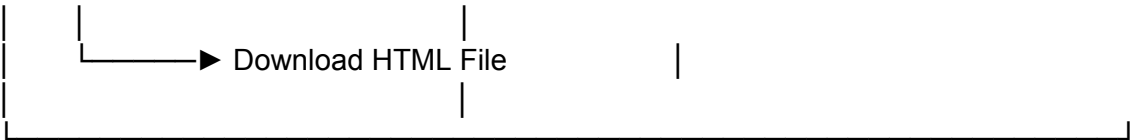
ENV-03: Storage Requirements

- Browser localStorage space (if used): ~5MB
- Downloaded file storage: User's disk space

7. Appendices

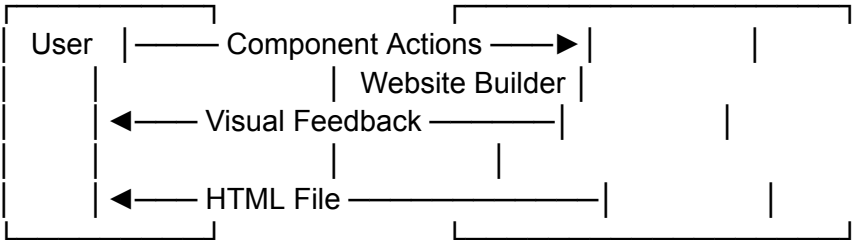
7.1 Appendix A: Use Case Diagrams



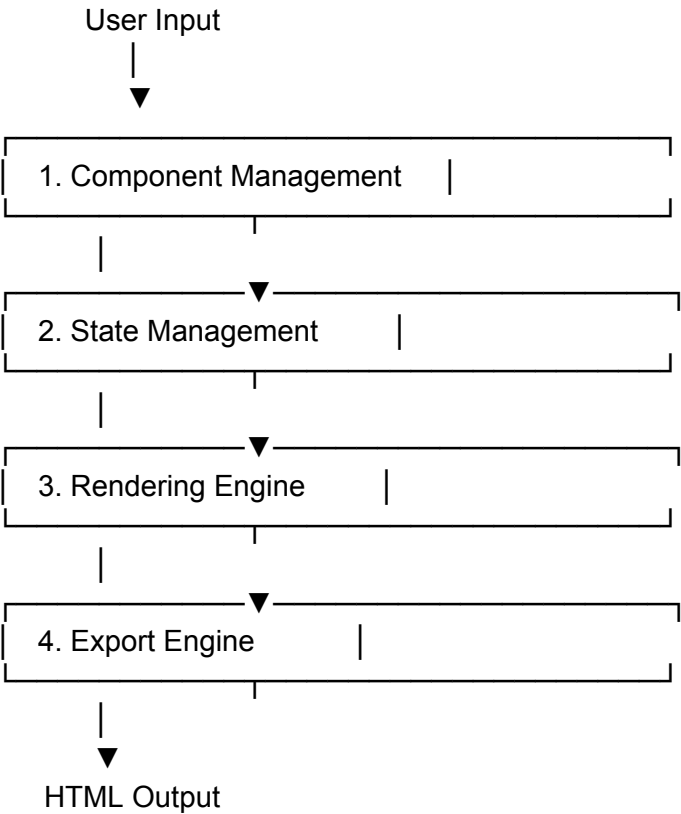


7.2 Appendix B: Data Flow Diagrams

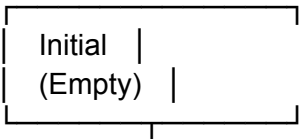
Level 0: Context Diagram

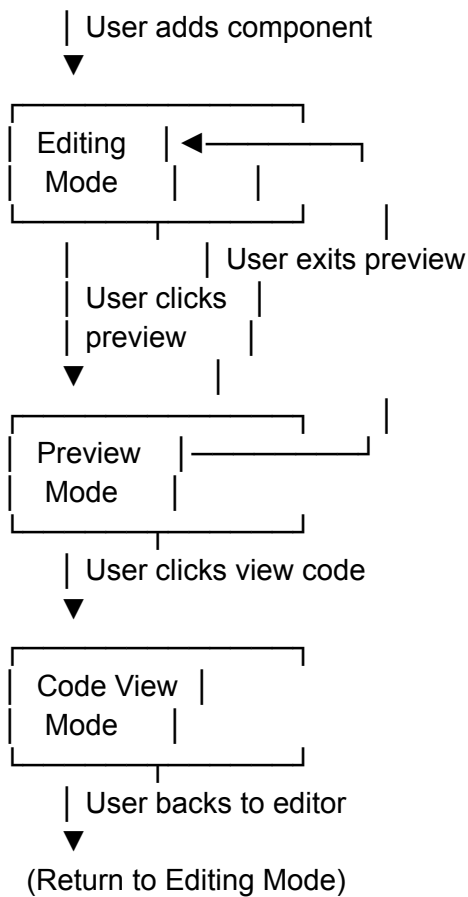


Level 1: Main Processes



7.3 Appendix C: State Transition Diagrams





7.4 Appendix D: Component State Structure

```
{
  components: [
    {
      id: 1670345678901,      // Unique timestamp ID
      type: "header",        // Component type
      content: {              // Type-specific content
        title: "Welcome",
        subtitle: "Your site"
      }
    },
    {
      id: 1670345678902,
      type: "text",
      content: {
        text: "This is a paragraph..."
      }
    },
    {
      id: 1670345678903,
      type: "image",
      content: {
```

```

        url: "https://example.com/image.jpg",
        alt: "Description",
        caption: "Image caption"
    },
    {
        id: 1670345678904,
        type: "button",
        content: {
            text: "Click Me",
            link: "https://example.com"
        }
    },
    {
        id: 1670345678905,
        type: "section",
        content: {
            col1: "Left column content",
            col2: "Right column content"
        }
    }
],
editingId: null,           // ID of component being edited
previewMode: false,       // Preview mode flag
showCode: false           // Code view flag
}

```

7.5 Appendix E: Testing Requirements

Unit Testing

- Component rendering functions
- State management functions
- HTML generation function
- Drag & drop handlers
- Input validation

Integration Testing

- Complete user workflows
- Component addition and deletion
- Edit-save-preview cycle
- Export and download process

Browser Compatibility Testing

- Chrome (Windows, macOS, Linux)
- Firefox (Windows, macOS, Linux)

- Safari (macOS, iOS)
- Edge (Windows)

Accessibility Testing

- Keyboard navigation
- Screen reader compatibility
- Color contrast validation
- WCAG 2.1 automated testing

Performance Testing

- Load time measurement
- Interaction responsiveness
- Large page performance (100 components)
- Memory usage profiling

7.6 Appendix F: Future Enhancements

Phase 2 Features:

- Save/Load projects (localStorage)
- Undo/Redo functionality
- Image upload (base64 encoding)
- Custom color scheme editor
- Font selection

Phase 3 Features:

- Template library (pre-built pages)
- Cloud save/sync
- Collaboration features
- Advanced components (forms, galleries)
- Animation options

Phase 4 Features:

- Mobile app version
- WordPress export plugin
- CMS integration
- A/B testing tools
- SEO optimization tools

7.7 Appendix G: Glossary

Canvas: The main editing area where components are placed and arranged

Component: A reusable UI building block (header, text, image, button, etc.)

Artifact: The final exported HTML file generated by the system

WYSIWYG: What You See Is What You Get - editing paradigm where content appears during editing as it will when displayed

Inline Editing: Editing content directly in place rather than in separate fields

State: The current data and configuration of the application

Drag Handle: Visual element used to initiate drag & drop operations

Export: Process of generating and downloading HTML file

Static Site: Website consisting of fixed HTML files (no server-side processing)

Document Approval

Role	Name	Signature	Date
Product Owner			
Development Lead			
QA Lead			
Stakeholder			

End of Software Requirements Specification

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